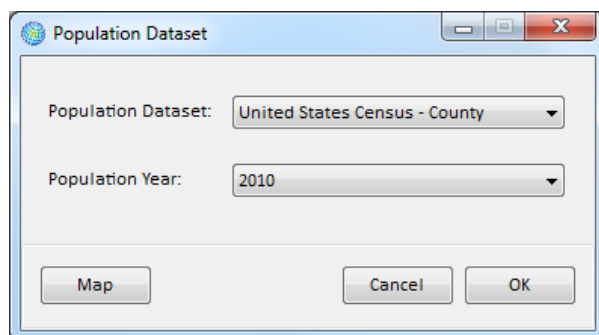


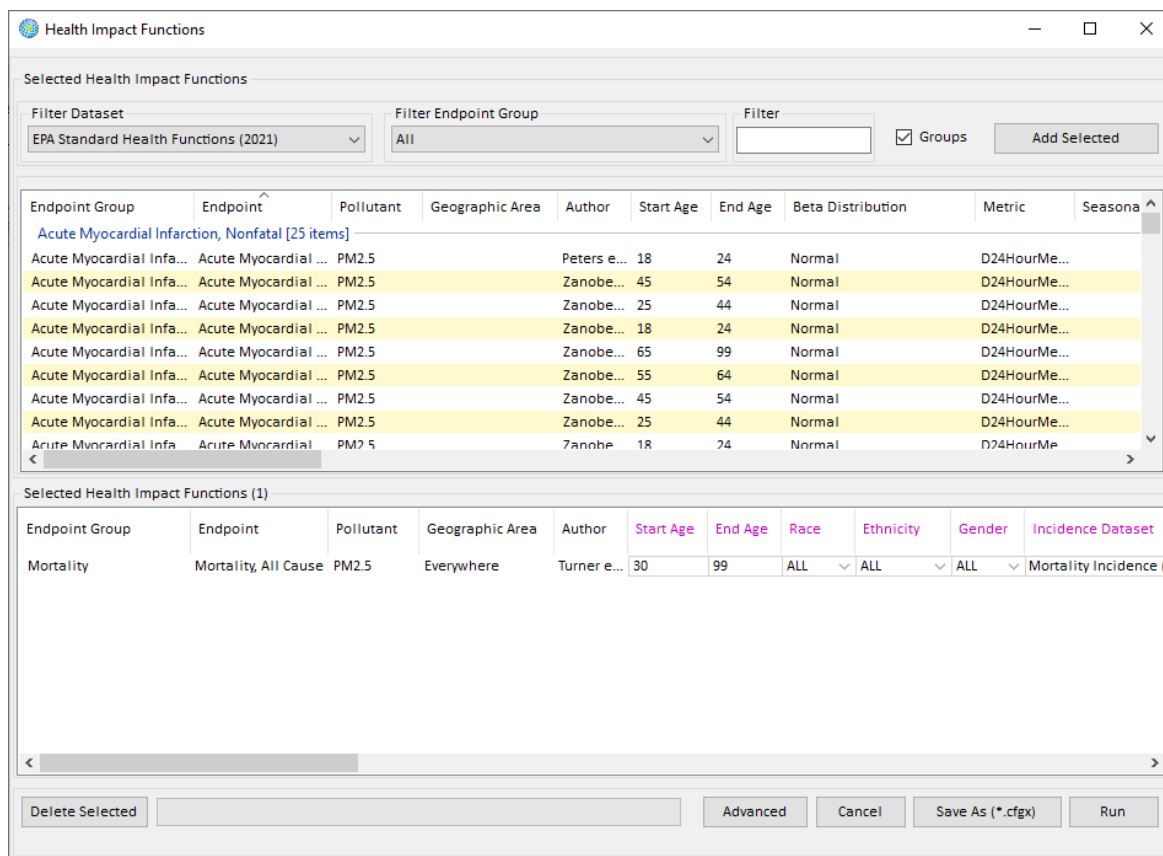
3.1.2 Estimate Health Impacts

The **Estimate Health Impacts** section allows you to calculate the change in the incidence of adverse health effects associated with changes in air quality. There are three steps in the process. The following is a brief description of each step. For detailed instructions, see Chapter 6: Estimating Incidence.

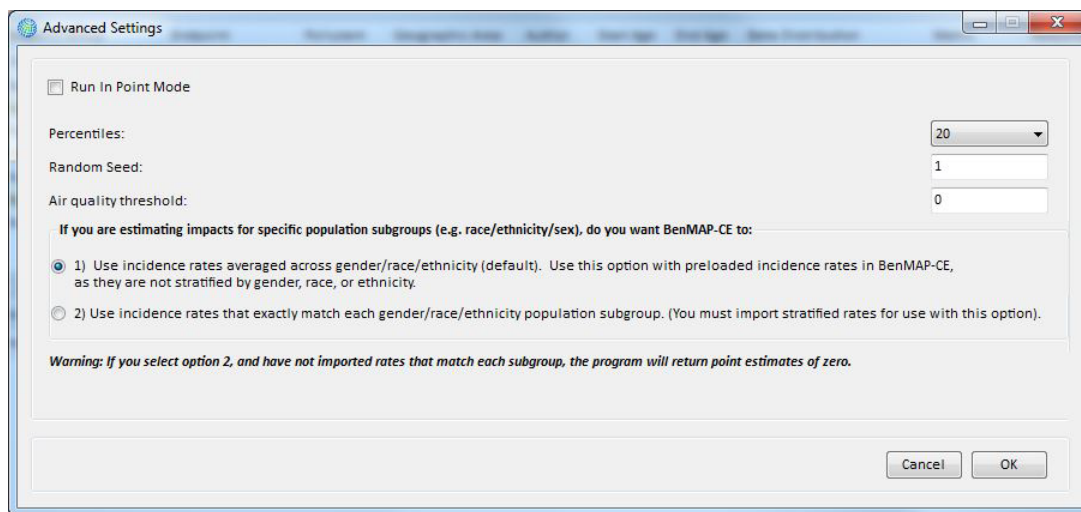
- **Step 1.** Specify the *Population Dataset* and *Population Year*.



- **Step 2.** Choose the *Health Impact Functions* that will be used to estimate the incidence of adverse health effects.



- **Step 3.** BenMAP-CE performs a full Monte-Carlo analysis to quantify the confidence intervals around mean incidence and economic value estimates by randomly sampling an uncertainty distribution around the effect coefficients or willingness to pay estimates. This step typically occurs automatically using default parameters for the Monte-Carlo analysis. However, you may change these defaults in Advanced Settings, as shown in the screenshot below. In general, the computation time increases as you specify additional percentiles to report from the Monte-Carlo generated distribution. If you want to replicate the Monte-Carlo distribution from another analysis, then you may also specify the *Random Seed*.) Specify the *Air Quality Threshold*, or a lowest value for air quality data. Any observations which fall below this threshold will be replaced with the threshold value in all calculations.



BenMAP-CE can store configuration choices in a user-named file with a *.cfgx* extension, and can store incidence change estimates in a user-named file with a *.cfgrx* extension.

BenMAP-CE Decision Point

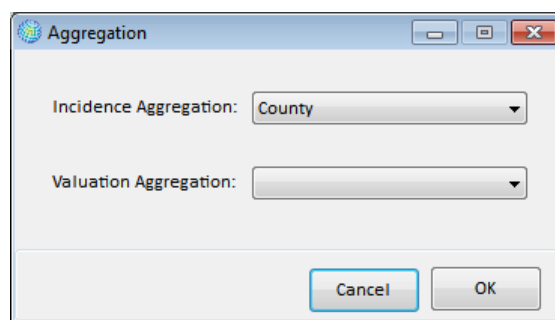
Once you generate a CFGRX file, BenMAP-CE will have quantified endpoint-specific incidence results, expressed as numbers of pollution-attributable cases, for each of the individual health impact functions you specified. At this point, you may:

1. Export your results to complete your analysis. (Proceed to Section 3.3.1)
2. Continue to the Aggregate, Pool, and Value part of the analysis (Section 3.1.3). Here you may:
 - a. Pool (i.e., combine) your incidence results for one or more health endpoints. Pooling is useful when you have multiple valid health impact functions for an endpoint that generate different results or have multiple subcategories of effects within a single endpoint that can be combined. (See Section 7.2.2)
 - b. Estimate the economic value of your incidence changes by applying health endpoint-specific valuation functions to those results. (See Section 7.2.3)

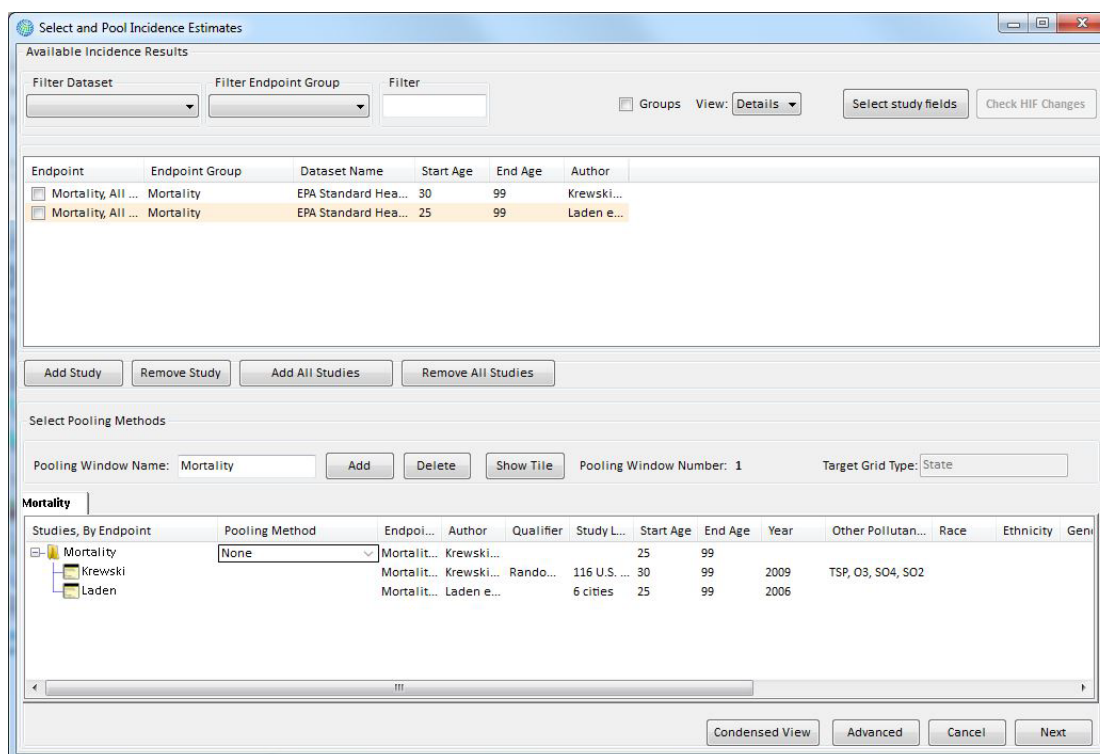
3.1.3 Aggregate, Pool, and Value

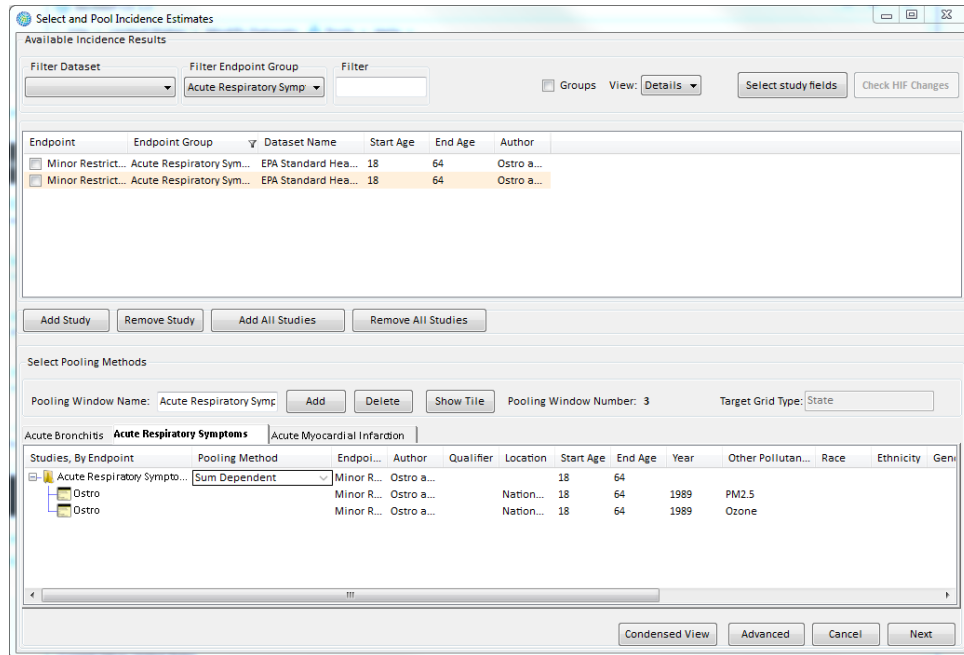
The **Aggregate, Pool, & Value (APV)** feature on the BenMAP-CE tree menu allows you to aggregate and pool previously calculated incidence estimates and place an economic value on these pooled and aggregated incidence estimates. You can also aggregate the economic values, and finally pool the aggregated economic values. There are several steps in this process. The following is a brief description of each step. For detailed instructions, see Chapter 7: Aggregating, Pooling, and Valuing.

- **Step 1.** Decide how to aggregate your results.

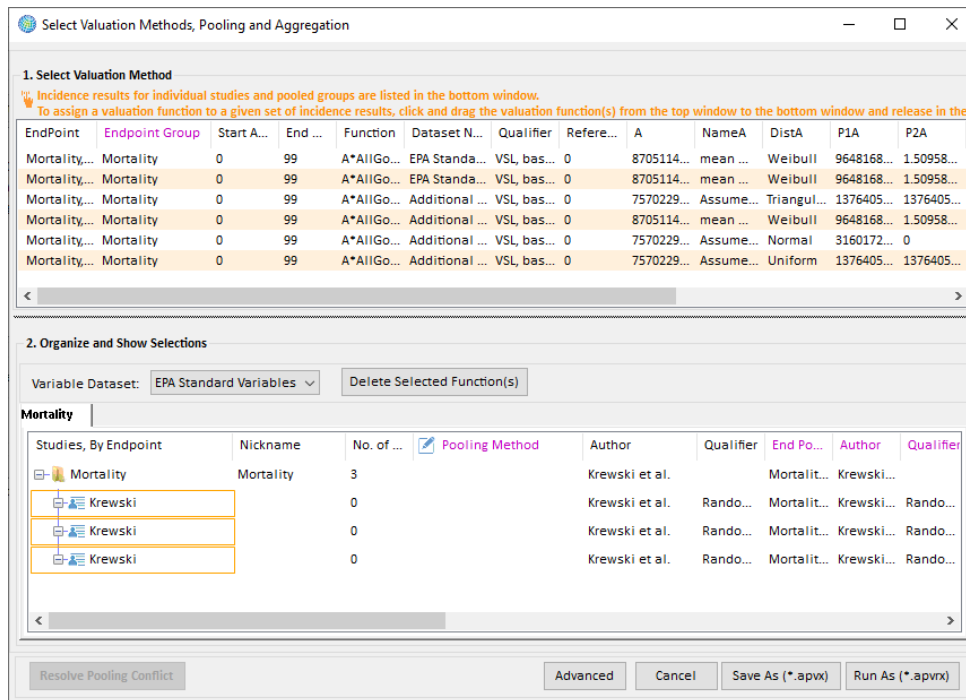


- **Step 2.** Decide how to pool and aggregate your incidence results.





- **Step 3.** Decide how you will estimate the economic value of your pooled and aggregated incidence results.



BenMAP-CE can store APV configuration choices in a user-named file with an *.apvx* extension, and can store APV configuration results in a user-named file with an *.apvrx* extension. You can access both files to use later.